

PARALLAX

Complete pipeline and models — distributed passive-sensor threat fusion platform

This document walks the full path from a raw acoustic/optical detection at one sensor node through to a fused, ranged, confidence-scored track on the command dashboard, and names every model or algorithm used at each stage. It is a software prototype: every waveform is a physically-motivated synthetic model and every accuracy figure is either derived from stated physics or marked as an estimate.

Pipeline at a glance

Stage	Module	What happens
1. Sensing	sim/edge_node.py	6-mic array capture, bandpass + energy-threshold onset detection, optional optical/IR fl
2. Bearing (DoA)	parallax/doa.py	GCC-PHAT time-delay estimation on all mic pairs + weighted least-squares plane-wave
3. Classification	parallax/features.py, classifier.py	24 physical features -> calibrated gradient-boosted-tree classifier
4. Wire report	parallax/contact.py	42-byte binary contact report (bearing, class, confidence, timestamp, CRC-16)
5. Fusion	parallax/fusion.py	Association, ranging (dt / triangulation / bearing-only), confidence scoring
6. Mission profile	parallax/profiles.py	Config-only retuning of weights & thresholds (perimeter / convoy / patrol / urban)
7. Visualization	viz/radar.py	Renders fused track JSON as a radar-style C2 dashboard PNG

Detailed pipeline

1. Per-node sensing

A 6-channel microphone array is sampled at 48 kHz, continuously DC-blocked and bandpass filtered 200 Hz–3 kHz for the direction-of-arrival branch only (the classifier sees the unfiltered full band, since filtering would discard the high-frequency structure that separates a gunshot from a firecracker). A rolling energy detector fires an onset when the signal exceeds 6σ of the tracked noise floor (threshold is mission-profile-tunable). A 200 ms pre-trigger ring buffer guarantees the transient's leading edge is captured rather than clipped. An optional optical/IR channel independently flags a muzzle flash, essentially instantaneous since light crosses 400 m in 1.3 ns.

2. Direction-of-arrival (bearing) estimation

Algorithm: GCC-PHAT (Generalized Cross-Correlation with Phase Transform), computed on all 15 pairs of the 6-element array. PHAT whitens the cross-spectrum, keeping only phase — this is what makes the correlation peak robust to reverberant reflections. Each pair yields a sub-sample time delay via parabolic interpolation of the correlation peak. Every delay becomes one linear equation in the source's unit direction vector u ; all 15 equations are stacked and solved by **weighted least squares** — closed-form, no iteration, no local minima, which is exactly what runs deterministically on an FPGA.

The array geometry is 5 microphones in a ring plus 1 raised on a short mast, not a flat ring — a planar array's fit is rank-deficient in the vertical axis, so the code explicitly refuses to report elevation from it rather than inventing a value. Bearing uncertainty (σ) is derived, not asserted: it combines the least-squares fit residual (how mutually inconsistent the 15 delays are) with the GCC-PHAT peak sharpness (how cleanly the correlation peak stands out from competing peaks). A weak or double peak sets a multipath-suspect flag and inflates σ automatically.

3. Transient classification

24 hand-crafted physical features are extracted from the full-band transient window: envelope shape (rise/decay time, duration, crest factor, kurtosis, skewness), spectral shape (centroid, spread, rolloff, flatness, slope, four sub-band energy ratios), harmonicity (tonal vs. impulsive), and low-frequency amplitude modulation (captures rotor blade-pass / engine firing-rate periodicity).

Model: HistGradientBoostingClassifier (scikit-learn's histogram-based gradient boosted decision trees) — max depth 4, 200 boosting iterations, learning rate 0.08, wrapped in **isotonic-regression probability calibration** (3-fold cross-validated) so that the confidence number consumed downstream by the fusion layer is actually meaningful rather than an overconfident raw tree vote. It classifies into five labels: GUNSHOT, VEHICLE, DRONE, PERSONNEL, NUISANCE, with NUISANCE (firecracker, door slam, thunder) treated as a first-class label rather than a generic reject bin.

A deep network (CNN over spectrograms) was deliberately not used: with only a few hundred realistic labeled examples realistically available, gradient-boosted trees on physically-motivated features generalize better in this low-data regime, train in seconds, run on an MCU, and every feature can be individually named and defended. Training uses a synthetic corpus generated by running the same physical scenario renderer (Friedlander blast wave + spherical spreading + atmospheric absorption) and the same windowing the live edge pipeline uses, specifically to avoid a train/inference domain gap.

4. Wire contact report

A node never transmits raw audio. It transmits a fixed **42-byte binary record** (bearing, bearing σ , threat class, confidence, event timestamp, node position, status flags, CRC-16/CCITT). One second of 6-channel 48 kHz/24-bit audio costs roughly 20,600× more link budget than one contact report — the reason edge classification is

non-negotiable on a radio-constrained mesh.

5. Fusion engine

The technical core of the system, and deliberately **not a learned model**: an explicit, physics-derived decision pipeline with four stages.

Time sync. Bearings are computed locally on each node's own clock, so mesh clock offset only needs to be good enough to associate events (~10 ms), not to align microsecond-scale delays across the mesh.

Association. Reports are clustered into a shared event using a physically-derived time window — flash↔acoustic bounded by $\text{max range} / c$; acoustic↔acoustic between nodes bounded by their baseline / c — plus a geometric plausibility check that the reported bearing rays actually cross in front of both nodes.

Ranging. Tried in priority order, never fabricated: (a) flash/acoustic Δt on a single node, $R = c \cdot \Delta t$, using measured air temperature for c ; (b) weighted least-squares triangulation of ≥ 2 bearing rays, weighted by $1/\sigma^2$, with robust outlier rejection when ≥ 3 bearings disagree; (c) bearing-only, reporting an honest ray rather than a fabricated position when neither applies.

Confidence scoring. An explicit weighted product of named, bounded terms: best single-report classifier confidence, modality corroboration (independent sensor types agreeing is stronger evidence than duplicates), spatial corroboration (independent nodes agreeing), geometry quality (tight triangulated fix vs. loose ray), and a multipath penalty. Deliberately not learned, since there is no labeled multi-node fusion corpus to fit a combiner to honestly.

6. Mission profiles

Four named configurations — perimeter, convoy, patrol, urban — that only retune sensor weights, detection thresholds, and bearing-sigma inflation. Same hardware, same edge pipeline, same fusion algorithm, same wire format in every profile.

7. Visualization

A separate, algorithm-agnostic renderer reads the fused-track JSON and draws a radar-style dashboard: bearing dial with a shaded 1σ uncertainty wedge, tracked position with its error ellipse, and disposition (e.g. ALERT). It knows nothing about signals, so it can be re-skinned without touching anything that computes a bearing.

Models and algorithms — summary

Component	Model / algorithm	Learned from data?
Bearing (direction-of-arrival)	GCC-PHAT cross-correlation + weighted least-squares plane-wave fit	No — closed-form signal processing / geometry
Transient classifier	HistGradientBoostingClassifier (scikit-learn) + isotonic probability calibration	Yes — trained on a synthetic corpus
Ranging	Physics formula $R = c \cdot \Delta t$ (flash/acoustic) or weighted least-squares triangulation	No — closed-form physics / geometry
Fusion confidence score	Explicit weighted product of named, bounded terms	No — hand-designed heuristic, by design

Classifier details

- Base estimator: HistGradientBoostingClassifier(max_depth=4, max_iter=200, learning_rate=0.08, l2_regularization=1.0)
- Calibration: CalibratedClassifierCV(method='isotonic', cv=3)
- Evaluation: 5-fold stratified cross-validation; reports per-class precision/recall and a confusion matrix rather than a single accuracy figure (accuracy on an imbalanced five-class problem flatters and informs nobody)
- Feature importance: permutation importance (5 repeats) reported per feature, so any prediction can be traced to a named physical property
- Input: 24-dimensional feature vector (see Feature set below)
- Output classes: GUNSHOT, VEHICLE, DRONE, PERSONNEL, NUISANCE

Feature set (24 features)

Temporal envelope: rise_time_ms, decay_time_ms, duration_ms, crest_factor_db, kurtosis, skewness, zero_crossing_rate, onset_slope_db_per_ms, decay_linearity, n_secondary_peaks, secondary_peak_ratio, temporal_centroid

Spectral shape: spectral_centroid_hz, spectral_spread_hz, spectral_rolloff95_hz, spectral_flatness, spectral_slope, energy_entropy

Sub-band energy ratios: band_ratio_lo (20–200 Hz, blast overpressure tail), band_ratio_mid (200–1500 Hz, muzzle blast core), band_ratio_hi (1.5–6 kHz, mechanical action / case ejection), band_ratio_vhi (6–20 kHz, crack / firecracker)

Periodicity: harmonicity (tonal vs. impulsive), modulation_20_200hz (blade-pass / cylinder-firing rate)

Wire format (42 bytes)

```
node_id (u16) | seq (u16) | t_event_ns (u64) | modality (u8) | threat (u8) | class_conf (u8) |
azimuth (u16, centidegrees) | azimuth_sigma (u8, decidegrees) | elevation (i16, centidegrees) |
elevation_sigma (u8) | range_m (u16) | range_sigma_m (u16) | peak_spl_db (u8) | snr_db (u8) | lat
(i32, 1e-7 deg) | lon (i32, 1e-7 deg) | alt_m (i16) | heading_deg (u16) | flags (u8) | crc16 (u16)
```

Generated from the PARALLAX repository source (parallax/, sim/, viz/). This is a software prototype: no hardware was built and no field recordings were made. See docs/04-limitations.md in the repository before quoting any figure.